Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/16/2006
PI Name:	Egolfopoulos, Fokion Ph.D.		
Project Title:	Quantitative Studies on the Propagation and Extinction of Near-Limit Premixed Flames Under Normal and Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Miller, Fletcher J	**Contact Phone:**	619-594-5791
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Spedding, Geoff (University of Southern California)		
Grant/Contract No.:	NNC04AA11A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To develop a variable pressure, counterflow burner apparatus equipped with several diagnostic instruments - PIV, PSV, LDV, thermocouples, and high-speed cinematography, to study weakly burning laminar premixed and diffusion flames in both normal and microgravity conditions. To obtain the quantitative information about the characteristics of weakly burning flames in microgravity through the use of the 2.2s drop tower, and use the information to analyze flame near-limit phenomena. Additionally, to use/develop detailed 1- and 2-D numerical models to simulate flame structures, and to validate proposed kinetics. Task Objective: To obtain quantitative information about the characteristics of weakly burning flames in microgravity, and to use this information to analyze near-limit phenomena at normal gravity and microgravity conditions. The research will allow for the study of weakly burning flames without the complications of buoyancy, wall heat loss or time varying strain. It will		

	provide fundamental data about the flammability limits, and has applications to turbulent combustion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It will provide fundamental data about the flammability limits, and has applications to turbulent combustion and space fire safety.
Task Progress:	In FY06, our research effort focused on (1) a new approach for the determining laminar flame speeds in a convenient manner especially under microgravity conditions, and (2) on the numerical determination of lean flammability limits under engine-like conditions. Laminar flame speeds have been widely used to (partially) validate chemical kinetics. Experimental techniques for determining laminar flame speeds include among others: the Bunsen flame, the counterflow, the porous burner, the spherical flame, and the transition from Bunsen to flat flame techniques. With the exception of the transition one, all these approaches need either extrapolation or extensive data processing. The transition technique on the other hand requires a very special set of conditions that allow for stable flame transition, and such conditions may not be met during the short times available at reduced gravity facilities. In the present study, the “Cake-Flame Technique” is introduced, which allows for stabilizing flame at ultra-low strain rates and subsequently measuring its propagation speed. It was found that the propagation speeds of cake flames are rather close to the laminar flame speeds determined by the counterflow technique after invoking appropriate extrapolations. The strain rates of cake flames were found to be very low, with attendant Karlovitz numbers of the order 10^-2 ~ 10^-3. Compared to other techniques, the cake flame one is much simpler and can be performed in short times. The determination of lean flammability limits under engine-like conditions were determined numerically using detailed kinetics and it was shown that a recent proposition that relates fundamental flammability limits to failure of engine operation below certain temperature is invalid.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Dong Y, Holley AT, Andac MG, Egolfopoulos FN, Davis SG, Middha P, Wang H. "Extinction of premixed H2/air flames: Chemical kinetics and molecular diffusion effects." Combustion and Flame. 2005 Sep;142(4):374-87. http://dx.doi.org/10.1016/j.combustflame.2005.03.017 , Sep-2005
Articles in Peer-reviewed Journals	Egolfopoulos FN, Holley AT, Law CK. "An Assessment of the Lean Flammability Limits of CH4/Air and C3H8/Air Mixtures at Engine-Like Conditions." Proceedings of the Combustion Institute 31, In press, December 2006. , Dec-2006